[3, 5, 4, 2, 9],   K = 3.

Sorting-Based: [2, 3, 4, 5, 9].

TC: O(n log n).
AS: O(1).

Max-Heap Based:

TC: O(n log n + K log n)

AS: O(n).

Min-Heap Based:

Kth largest == smallest of the K-largest elements

[3, 5, 4, 2, 9].

K=3.

Min-Heap.

~~3~~, 5, ④ 9

TC: O(n log K).
AS: O(K)

$$\left[\begin{array}{cccc} \cancel{1} & \cancel{3} & 5 & 10 \\ \cancel{2} & 3 & 3 & 9 \\ \cancel{1} & 6 & 11 & 13 \\ 12 & 12 & 14 & 15 \end{array}\right]$$

o/p: 1, 1, 2, 3, 3, 3, 5, 6, 9, 10, 11, 12, 12, 13, 14, 15.

[1, 1, 2, 3 . . . . . .].

~~1~~, ~~2~~, ~~1~~, 12, 3, 6, 3 . . .

2K + 3K + 4K + . . . K.K

K (2+3+ . . . . K) = O(K³).

– (1, 2, 1, 12)

$O(K \cdot K^2) = O(K^3)$.

P.Q. Based

TC: $O(K^2 \log K)$

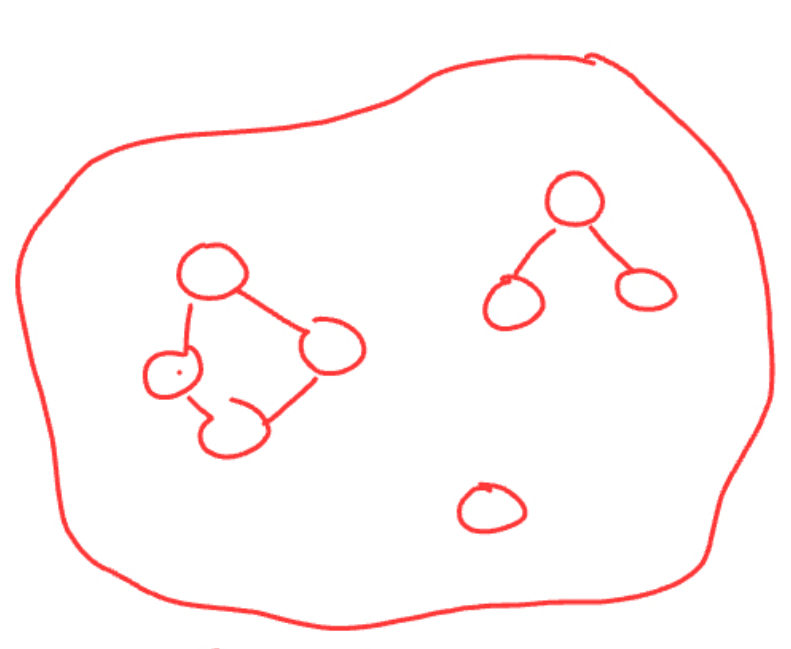
AS: O(K).

{val, (i, j)}.

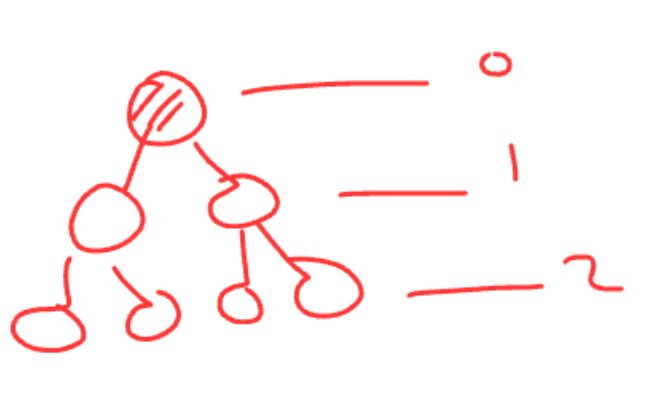
$\log(K^2)$
$= 2 \log K$

1, 2, 3.

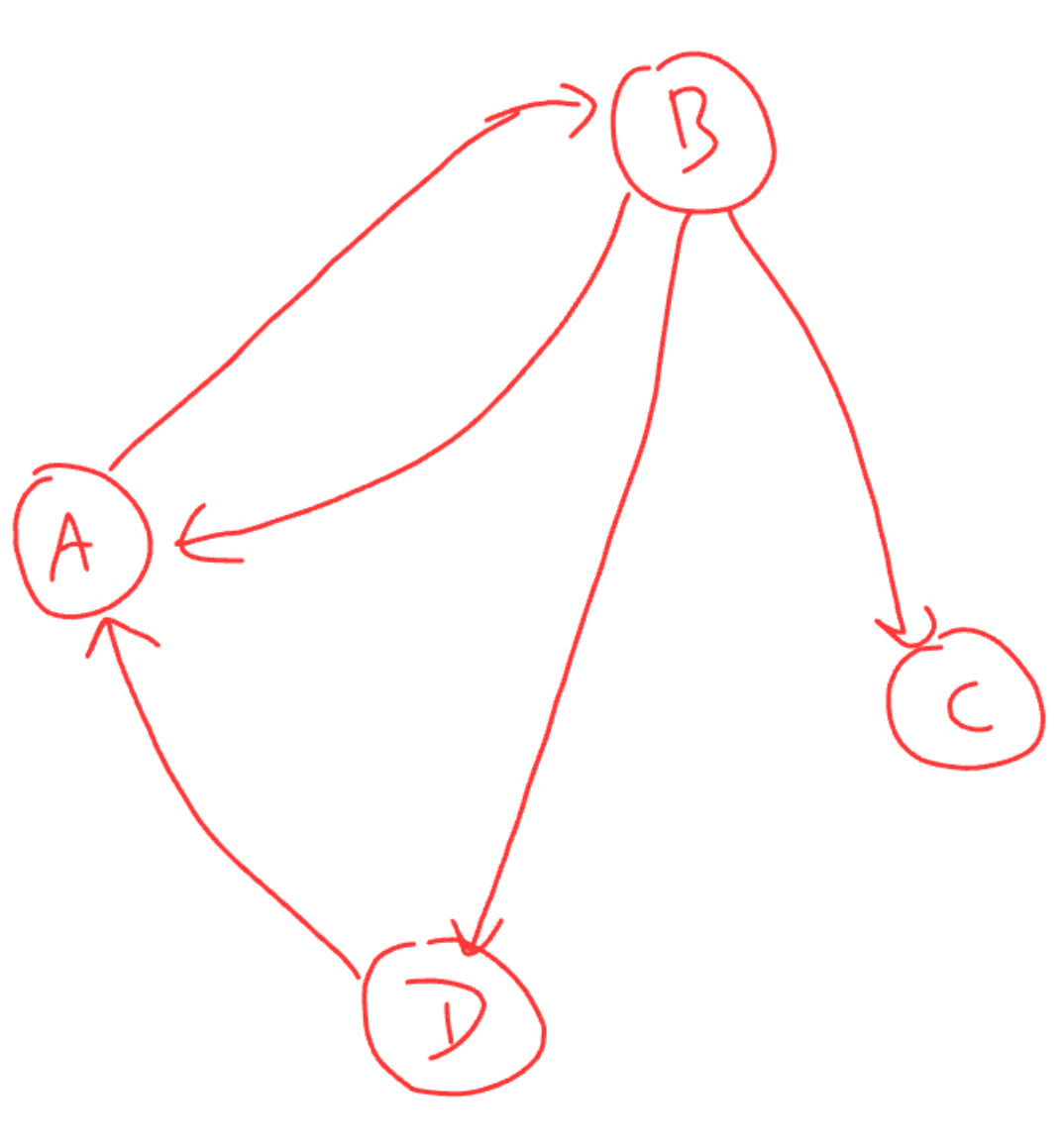
–1, –2, –3



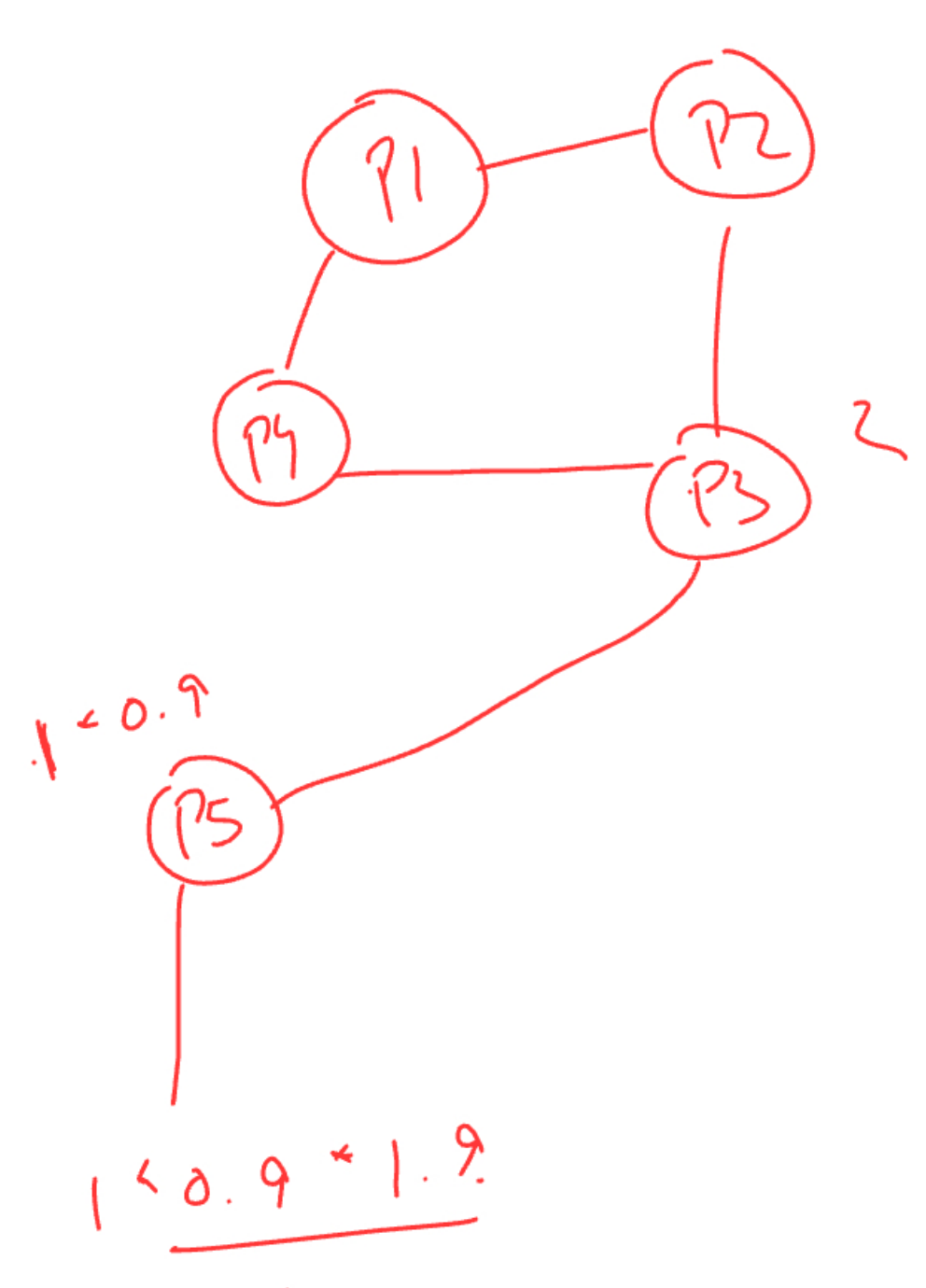
Graph.



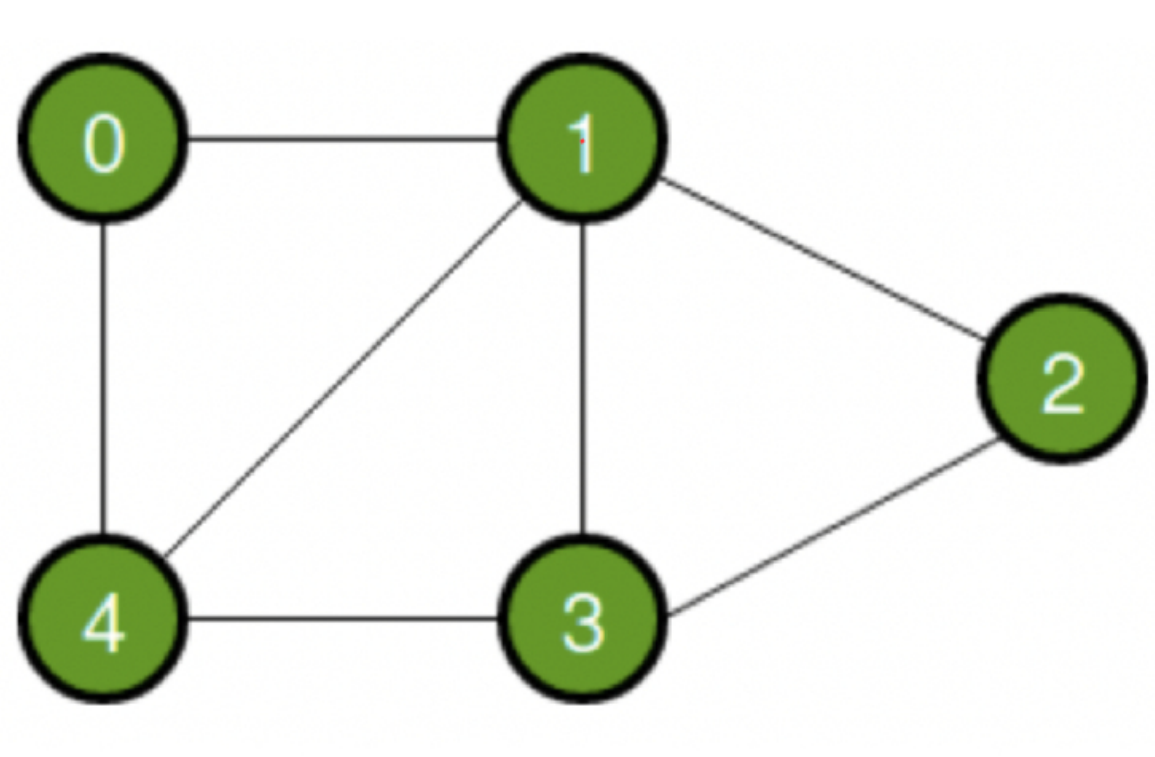
Tree.



0.1

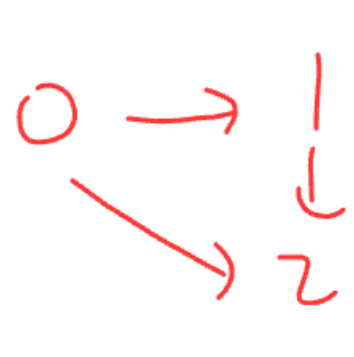


Adj. Matrix



$$\begin{array}{c|ccccc} & 0 & 1 & 2 & 3 & 4 \\ \hline 0 & 0 & 1 & 0 & 0 & 1 \\ 1 & 1 & 0 & 1 & 1 & 1 \\ 2 & 0 & 1 & 0 & 1 & 0 \\ 3 & 0 & 1 & 1 & 0 & 1 \\ 4 & 1 & 1 & 0 & 1 & 0 \end{array}$$

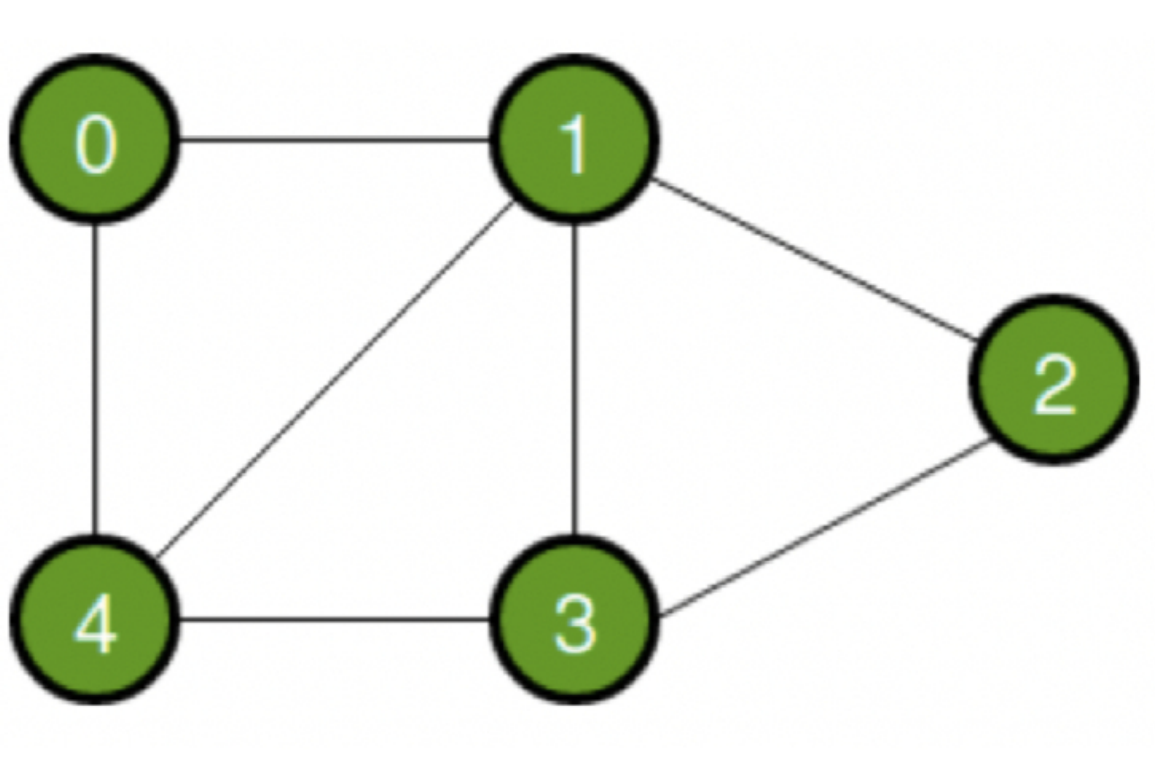
Space: O(V²).



$$\begin{array}{c|ccc} & 0 & 1 & 2 \\ \hline 0 & 0 & 1 & 1 \\ 1 & 0 & 0 & 1 \\ 2 & 0 & 0 & 0 \end{array}$$

adj[i][j] = w   if $i \xrightarrow{w} j$

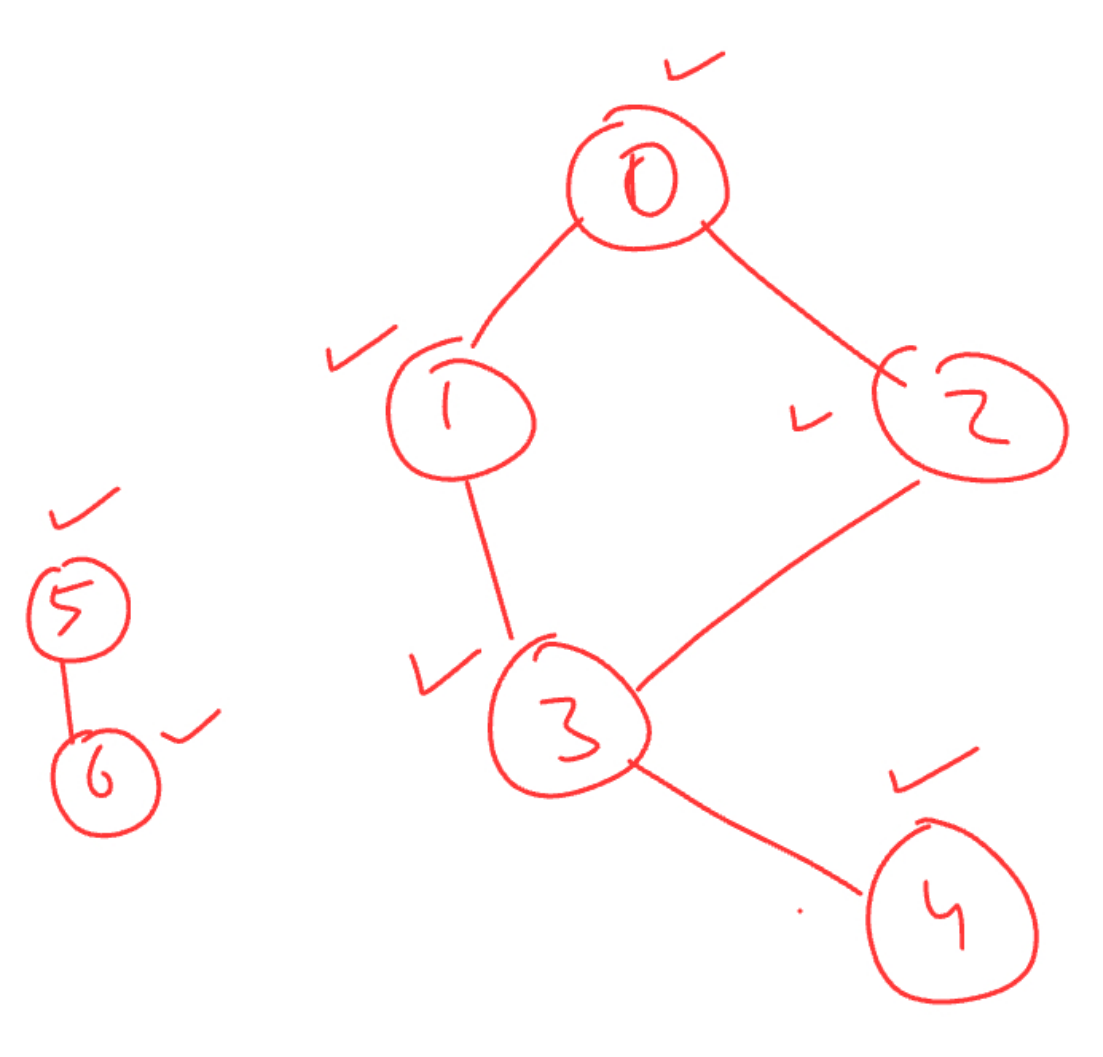
Adj list.



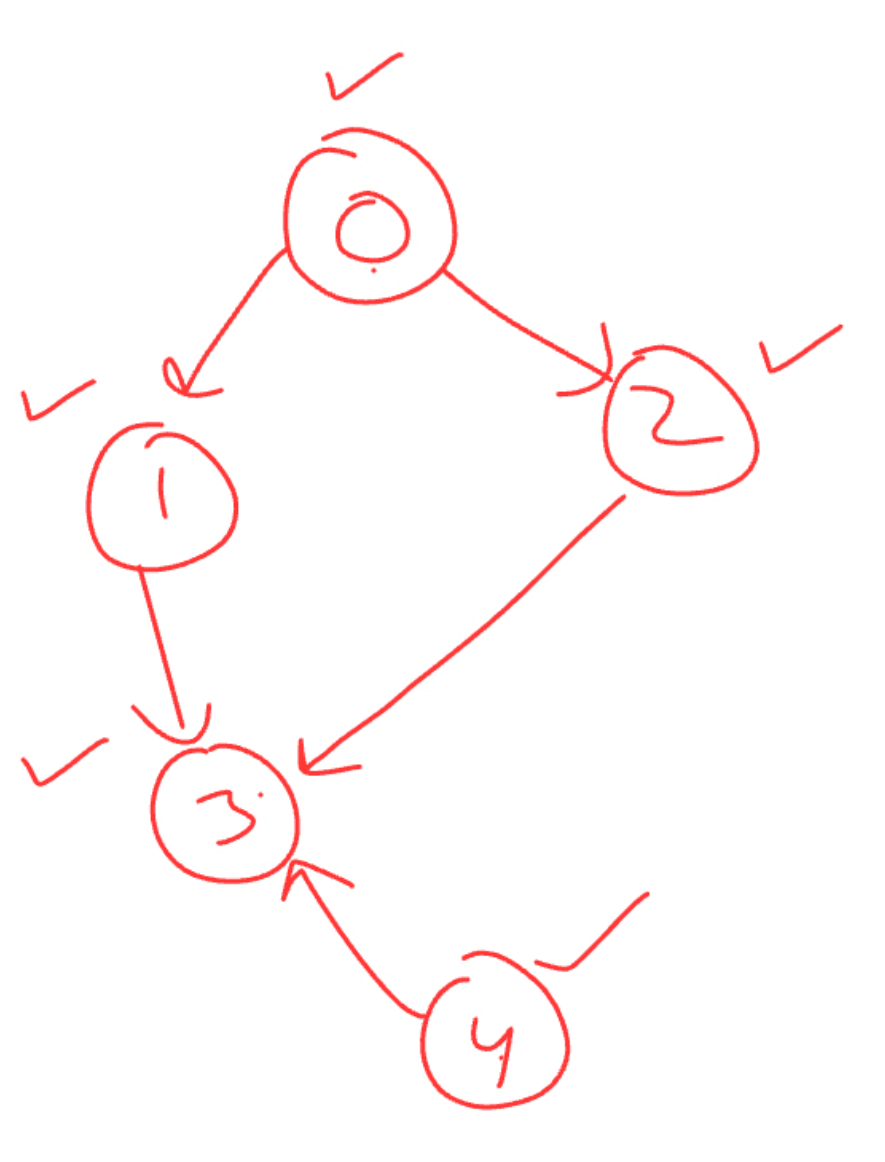
0: [1, 4]
1: [0, 4, 3, 2]
2: [1, 3]
3: [4, 1, 2]
4: [0, 1, 3].

Space = O(V + E)

DFS.



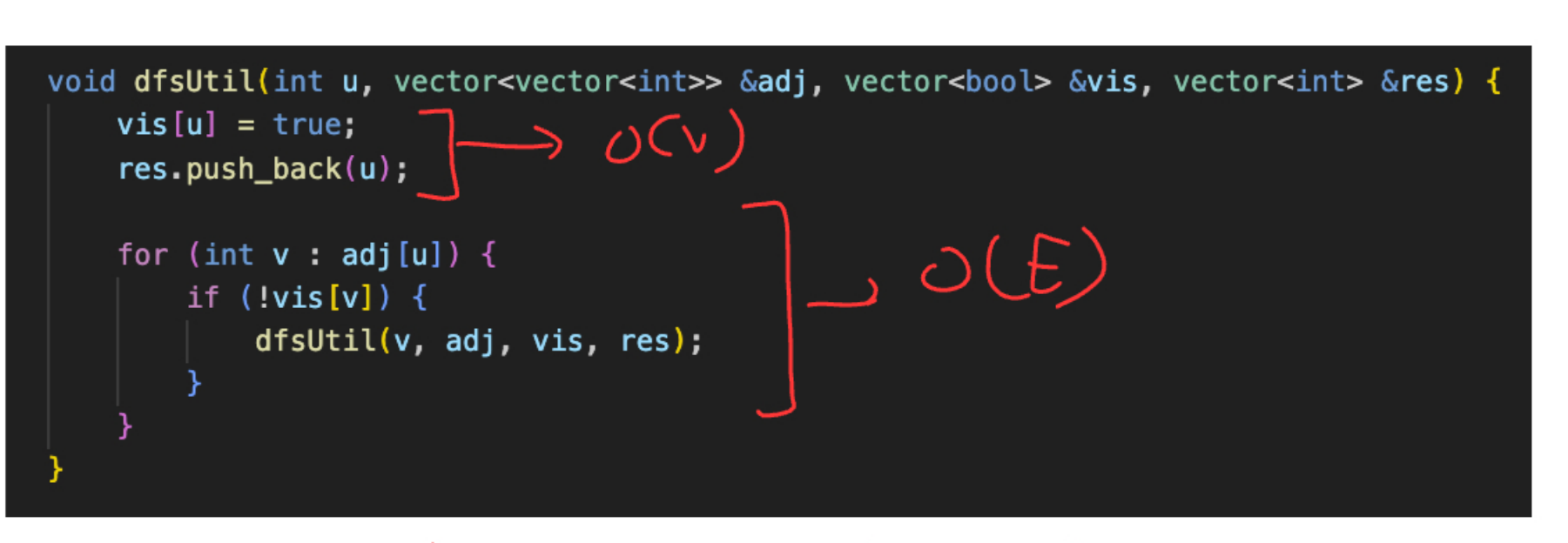
[0, 1, 3, 2, 4, 5, 6].



[0, 1, 3, 2, 4].

e1, e2 . . . . . . , ev

e1 + e2 + e3 — — — . ev = E



```
void dfsUtil(int u, vector<vector<int>> &adj, vector<bool> &vis, vector<int> &res) {
    vis[u] = true;
    res.push_back(u);

    for (int v : adj[u]) {
        if (!vis[v]) {
            dfsUtil(v, adj, vis, res);
        }
    }
}
```

→ O(V)

→ O(E)

= O(V + E).